

Technology Stack

HDI Prediction System | Team Lead: Farzana | Contributors: Chinthoti Raghavendra, Ahmad Kabir Shaik, Naga Sireesha Naralasetty, Sitha Ramaraju Velisala

Overview

The HDI Prediction System is built using a Python-based machine learning stack, with a lightweight Flask web application for deployment and a simple, dark-themed front end for user interaction.

Core stack

Layer	Technology
Programming language	Python 3.14
Data processing	Pandas, NumPy
Machine learning	Scikit-learn (Linear Regression)
Model persistence	Pickle (.pkl file)
Backend framework	Flask
Frontend	HTML, CSS (dark-themed UI)
Development environment	Jupyter Notebook
Version control	Git & GitHub
Deployment	Render.com

Why these choices

- Scikit-learn's Linear Regression offers a simple, interpretable model well-suited to the near-linear relationship between HDI and its underlying indicators.
- Flask is lightweight and quick to deploy, ideal for a focused single-purpose prediction app.
- Render.com provides free-tier hosting suitable for an academic project demo link.
- Git and GitHub support version control and easy collaboration across the team.